

An Exact Classical–Quantum Coexistence Trade-off Through One Counted Qubit, With Covariant Attainability

The exact optimal trade-off between classical label accuracy and same-sector entanglement fidelity for one unassisted qubit under a $\mathbb{Z}_2$ symmetry, with a covariant attaining family, a nonunitarity transition, and a nonconvex attainable region.

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Abstract

We study a minimal *coexistence* task: a single qubit of memory must be used, through one unassisted encode–decode factorization, both to report a classical binary label (which of two orthogonal “sectors” a carrier occupied) and to preserve quantum information within one sector, in the presence of a $\mathbb{Z}_2$ symmetry. We quantify the first by a label accuracy L — which we show is exactly a two-state discrimination success probability for a decoder-induced measurement — and the second by the unconditional same-sector entanglement fidelity F_{same} . Our main result is the exact optimal trade-off: for every required accuracy $\tau \in [\frac{1}{2}, 1]$, the maximum attainable F_{same} is a single closed-form curve $\Phi(\tau)$, and this maximum is *the same whether or not the factorization is required to be covariant* under the symmetry. An explicit two-parameter covariant CPTP family attains Φ at every threshold, so covariance imposes no optimal-fidelity penalty anywhere on the trade-off. The frontier has a structural transition at $\tau = \frac{5}{8}$: at or below it a unitary code encoder can be optimal, strictly above it every optimal code-restricted encoder is necessarily nonunitary. The attainable score region is nonconvex, which makes the prohibition of an uncounted shared classical index between encoder and decoder mathematically consequential. We position the result against binary-discrimination/disturbance theory and one-shot hybrid coding; in particular a directly applicable hybrid-memory inequality (Kuperberg) specializes here to $F_{\text{same}} \leq (5 - 2\tau)/4$, which is valid but strictly loose for $\tau > \frac{1}{2}$ (e.g. $\frac{7}{8}$ against the sharp $\frac{1}{2}$ at $\tau = \frac{3}{4}$) and tight at $\tau = \frac{1}{2}$. Whether the exact frontier, the transition, or covariant optimality follow from earlier results remains open. The results are proved analytically, with selected exact symbolic identities, explicit constructions, and regression tests supporting the proofs.

1. Introduction

A recurring question in quantum information is how much a limited memory must *give up* on one figure of merit to do well on another when it is forced to serve two purposes at once. We isolate a sharp instance in which one figure is classical (correctly reporting a binary label) and the other is quantum (preserving entanglement fidelity), the memory is exactly one qubit, and a $\mathbb{Z}_2$ symmetry is present so that one can ask whether *respecting the symmetry* — a natural structural constraint — carries a cost.

The task, made precise in Section 2, fixes a four-dimensional carrier split into two orthogonal two-

dimensional *sectors* that are each *preserved* by a $\mathbb{Z}_2$ symmetry $U = X \otimes X$ (which permutes the two basis vectors within each sector). A single encoder $\mathcal{E}$ compresses the carrier into one qubit and a single decoder $\mathcal{D}$ expands it back, with no side resource. Two scores grade the composite channel $\mathcal{N} = \mathcal{D} \circ \mathcal{E}$: the **label accuracy** L , how reliably a fixed two-outcome readout of $\mathcal{N}$ distinguishes the two sectors, and the **same-sector entanglement fidelity** F_{same} , how well $\mathcal{N}$ preserves quantum information inside one fixed sector. One may additionally require $\mathcal{E}$ and $\mathcal{D}$ to be *factorwise covariant* — to intertwine the carrier symmetry with a representation of $\mathbb{Z}_2$ on the counted qubit.

We prove three things. First (Section 3–4), the optimal trade-off is exact and closed-form: writing $f_{\text{all}}(\tau)$ and $f_{\text{cov}}(\tau)$ for the largest F_{same} achievable at $L \geq \tau$ over all admissible, respectively all covariant, factorizations,

$$f_{\text{all}}(\tau) = f_{\text{cov}}(\tau) = \Phi(\tau) \quad \text{for all } \tau \in [\tfrac{1}{2}, 1].$$

Second (Section 5), an explicit covariant family attains Φ at every τ , so the equality above is realized by symmetry-respecting designs — **covariance is cost-free along the entire optimal trade-off**. Third (Section 6), the frontier has a genuine structural transition at $\tau = \frac{5}{8}$ and the attainable region is nonconvex. Section 7 places the result in the landscape of information–disturbance and hybrid-coding results, including a directly applicable but loose prior bound. Section 8 states scope precisely. Two appendices give an exact algebraic reformulation of the task as a hybrid memory and an inventory of the machine-checked identities.

We emphasize at the outset what is *not* claimed. The equality $f_{\text{all}} = f_{\text{cov}}$ is an equality of *optima at each threshold*, not of the full attainable regions; it does not assert that every unrestricted channel has a covariant counterpart matching both scores (every covariant channel is, trivially, already unrestricted-admissible). The frontier is stated for a fixed qubit representation of the symmetry, made explicit in Section 2; it is not representation-independent. The relationship to prior work is stated precisely, once, in Section 7.

2. The counted-resource model

Carrier, symmetry, code. The carrier is $\mathbb{C}^4$ with computational basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. The $\mathbb{Z}_2$ symmetry acts by $U = X \otimes X$. The *same sector* is the code subspace fixed by the isometry $W : \mathbb{C}^2 \rightarrow \mathbb{C}^4$, $W|0\rangle = |00\rangle$, $W|1\rangle = |11\rangle$; its projector is $Q_s = WW^\dagger = |00\rangle\langle 00| + |11\rangle\langle 11|$, and the *different sector* is $Q_d = \mathbb{1} - Q_s = |01\rangle\langle 01| + |10\rangle\langle 10|$. The symmetry preserves each sector but acts nontrivially on the code: $UW = WX$, i.e. U restricts to logical X on the same-sector qubit. The two orthogonal carrier mixtures $Q_s/2$ and $Q_d/2$ are the sector test inputs.

Resource and admissibility. The counted resource is one qubit $\mathbb{C}^2$. An *admissible* channel is a composite $\mathcal{N} = \mathcal{D} \circ \mathcal{E}$ with $\mathcal{E} : \mathcal{L}(\mathbb{C}^4) \rightarrow \mathcal{L}(\mathbb{C}^2)$ and $\mathcal{D} : \mathcal{L}(\mathbb{C}^2) \rightarrow \mathcal{L}(\mathbb{C}^4)$ both CPTP, and *nothing else*: no shared randomness, no side channel, no postselection, and in particular no uncounted classical index correlating $\mathcal{E}$ with $\mathcal{D}$. A *covariant* admissible channel additionally has factorwise-covariant factors: for the memory representation V of $\mathbb{Z}_2$ on the qubit,

$$\mathcal{E} \circ \text{Ad}_U = \text{Ad}_V \circ \mathcal{E}, \quad \mathcal{D} \circ \text{Ad}_V = \text{Ad}_U \circ \mathcal{D}.$$

Throughout, the memory representation in the covariant class is $V = Z$, the fixed $(1, 1)$ representation; replacing Z by $-Z$ leaves the conjugation action Ad_V unchanged, so V is the sign class $\{Z, -Z\}$. The frontier below is stated for this representation and is *not* representation-independent. With the trivial representation $V = \mathbb{1}$, for instance, encoder covariance forces

$T = T \circ \Delta_X$ where $\Delta_X(\rho) = \frac{1}{2}(\rho + X\rho X)$ is the X -basis measure-and-reprepare (pinching) map; hence the code-restricted encoder is entanglement-breaking, its recovered code map has rank-one Kraus operators K_j , and $F_{\text{same}} = \frac{1}{4} \sum_j |\text{Tr} K_j|^2 \leq \frac{1}{4} \sum_j \text{Tr}(K_j^\dagger K_j) \leq \frac{1}{2}$. An X -basis measure-and-reprepare strategy attains $(L, F_{\text{same}}) = (\frac{1}{2}, \frac{1}{2})$ with $V = \mathbb{1}$ -covariant factors, so the optimum at $\tau = \frac{1}{2}$ is $\frac{1}{2}$ rather than the value 1 of Theorem 1.

The two scores. Let $\mathcal{N} = \mathcal{D} \circ \mathcal{E}$ be admissible. Define the decoder-induced effect $A = \mathcal{D}^\dagger(Q_s)$ and the encoder outputs $\sigma_s = \mathcal{E}(Q_s/2)$, $\sigma_d = \mathcal{E}(Q_d/2)$. Because $\mathcal{D}$ is trace-preserving, $\mathcal{D}^\dagger$ is unital, so $A = \mathcal{D}^\dagger(Q_s)$ satisfies $0 \leq A \leq \mathbb{1}$ and $\{A, \mathbb{1} - A\}$ is a valid two-outcome POVM. The **label accuracy** is the average success probability of this readout in distinguishing the two sector inputs,

$$L = \frac{1}{2} \text{Tr}(A\sigma_s) + \frac{1}{2} \text{Tr}[(\mathbb{1} - A)\sigma_d].$$

By the adjoint identity $\text{Tr}(\mathcal{D}^\dagger(Q_s)\rho) = \text{Tr}(Q_s \mathcal{D}(\rho))$ this is exactly the labelling probability of the composite channel,

$$L = \frac{1}{2} [\text{Tr}(Q_s \mathcal{N}(Q_s/2)) + \text{Tr}(Q_d \mathcal{N}(Q_d/2))] = \frac{t_s + t_d}{2},$$

where $t_s = \text{Tr}(Q_s \mathcal{N}(Q_s/2))$ and $t_d = \text{Tr}(Q_d \mathcal{N}(Q_d/2))$ are the two correct-label probabilities. Thus L is a genuine equal-prior two-state discrimination success probability for a decoder-induced, jointly optimized measurement; the ordinary classical decoding error is $1 - L$. The **same-sector entanglement fidelity** grades quantum preservation inside the code,

$$F_{\text{same}} = \frac{1}{4} \sum_{a,b \in \{0,1\}} \langle Wa | \mathcal{N}(W|a) \langle b | W^\dagger | Wb \rangle,$$

i.e. the Schumacher entanglement fidelity of the code-restricted channel $R \mapsto W^\dagger \mathcal{N}(WRW^\dagger)W$, scored unconditionally on the one fixed sector.

The optimization. For $\bullet \in \{\text{all}, \text{cov}\}$ (the covariant class taken at $V = Z$) set

$$f_\bullet(\tau) = \max \{F_{\text{same}}(\mathcal{N}) : \mathcal{N} \text{ admissible, (covariant if } \bullet = \text{cov}), L(\mathcal{N}) \geq \tau\}.$$

The label accuracy obeys $0 \leq L \leq 1$, and *not* $L \geq \frac{1}{2}$: since the readout effect is fixed by the decoder, a label may be systematically reversed — an admissible Z -covariant “label-swap” attains $L = 0$ (Appendix B). The reason the interesting range is $\tau \in [\frac{1}{2}, 1]$ is instead that a covariant strategy attains $(L, F_{\text{same}}) = (\frac{1}{2}, 1)$ and no fidelity exceeds 1, so $f_{\text{cov}}(\tau) = f_{\text{all}}(\tau) = 1$ already for every $\tau \leq \frac{1}{2}$; the nontrivial trade-off begins at $\frac{1}{2}$.

3. Main result

Theorem 1 (covariant class at $V = Z$). For every $\tau \in [\frac{1}{2}, 1]$,

$$f_{\text{all}}(\tau) = f_{\text{cov}}(\tau) = \Phi(\tau) := \begin{cases} \frac{(1 + \sqrt{3 - 4\tau})^2}{4}, & \frac{1}{2} \leq \tau \leq \frac{5}{8}, \\ \frac{(1 + \sqrt{2 - \sqrt{4\tau - 2}})^2}{4}, & \frac{5}{8} \leq \tau \leq 1. \end{cases}$$

The two branches meet at $\tau = \frac{5}{8}$ with value $\frac{3+2\sqrt{2}}{8}$, where Φ is C^1 . In particular $\Phi(\frac{1}{2}) = 1$, $\Phi(\frac{3}{4}) = \frac{1}{2}$, $\Phi(1) = \frac{1}{4}$.

The proof has two halves: an upper bound $f_{\text{all}} \leq \Phi$ that assumes nothing about covariance (Section 4), and an explicit covariant family achieving $F_{\text{same}} = \Phi(\tau)$ at $L = \tau$ (Section 5), giving $f_{\text{cov}} \geq \Phi$. Since $f_{\text{cov}} \leq f_{\text{all}} \leq \Phi \leq f_{\text{cov}}$, all three coincide. The covariant lower bound is what forces the two optima to be equal: a symmetry-respecting design already reaches the unrestricted ceiling.

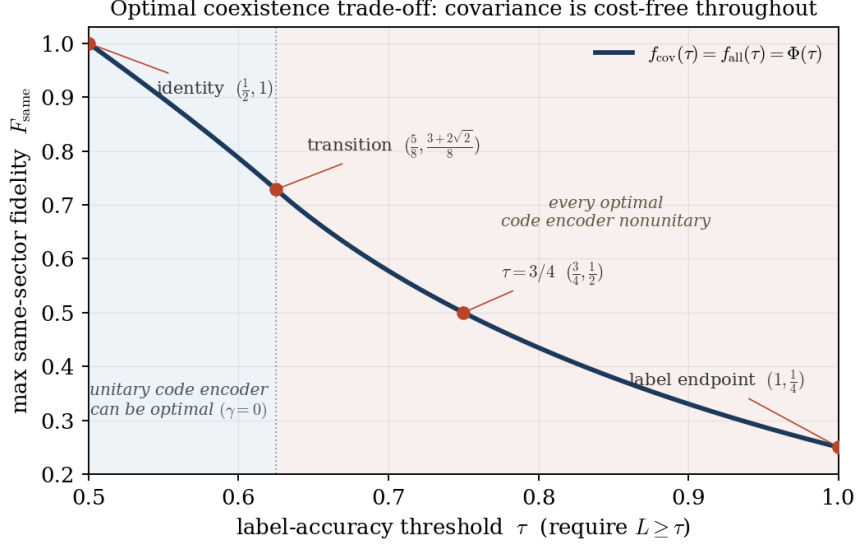


Figure 1: The optimal coexistence trade-off $\Phi(\tau)$. The curve is simultaneously the unrestricted optimum f_{all} and the covariant optimum f_{cov} ; they coincide, so there is no covariance gap. Marked are the exact points $(\frac{1}{2}, 1)$, $(\frac{5}{8}, \frac{3+2\sqrt{2}}{8})$, $(\frac{3}{4}, \frac{1}{2})$, $(1, \frac{1}{4})$ and the transition at $\tau = \frac{5}{8}$. Above $\frac{5}{8}$ every optimal code-restricted encoder is necessarily nonunitary.

4. The upper bound

Fix an admissible $\mathcal{N} = \mathcal{D} \circ \mathcal{E}$. Let $T(\rho) = \mathcal{E}(W\rho W^\dagger)$ be the *code-restricted encoder* (a qubit channel with Kraus operators E_μ) and $B(\tau) = W^\dagger \mathcal{D}(\tau)W$ the code-restricted decoder branch (CP, trace-nonincreasing, Kraus B_ν), so that $A = B^\dagger(\mathbb{1})$ and $F_{\text{same}} = \frac{1}{4} \sum_{\mu,\nu} |\text{Tr}(B_\nu E_\mu)|^2$.

Coordinates. Diagonalize the memory effect $A = \alpha \text{diag}(1, \eta)$ with $0 < \alpha \leq 1$, $0 \leq \eta \leq 1$ (the zero-effect case $A = 0$, i.e. $\alpha = 0$, gives $L = \frac{1}{2}$ and $F_{\text{same}} = 0$; it is *feasible* only at $\tau = \frac{1}{2}$ in the range under study and is not optimal there, since $\Phi(\frac{1}{2}) = 1$). Rescaling the decoder branch by $\tilde{B}_\nu = B_\nu / \sqrt{\alpha}$ gives $F_{\text{same}} = \alpha F'$ with $F' = \frac{1}{4} \sum |\text{Tr}(\tilde{B}_\nu E_\mu)|^2$. Independently diagonalize the logical input effect $G = T^\dagger(|0\rangle\langle 0|) = \text{diag}(a, b)$, $0 \leq b \leq a \leq 1$, by a logical basis change $E_\mu \mapsto E_\mu V$, $B_\nu \mapsto V^\dagger B_\nu$, which preserves every Kraus trace and hence A , σ_s , and F_{same} .

Coupled inequality. Writing $E_\mu = \begin{pmatrix} r_\mu & s_\mu \\ t_\mu & u_\mu \end{pmatrix}$ and $\tilde{B}_\nu = \begin{pmatrix} x_\nu & y_\nu \\ z_\nu & w_\nu \end{pmatrix}$, trace preservation of T and the diagonalization of G give the exact norm and orthogonality relations

$$\langle \mathbf{r}, \mathbf{s} \rangle = \langle \mathbf{t}, \mathbf{u} \rangle = 0, \quad \|\mathbf{r}\|^2 = a, \quad \|\mathbf{s}\|^2 = b, \quad \|\mathbf{t}\|^2 = 1-a, \quad \|\mathbf{u}\|^2 = 1-b, \quad \|\mathbf{x}\|^2 + \|\mathbf{z}\|^2 = 1, \quad \|\mathbf{y}\|^2 + \|\mathbf{w}\|^2 = \eta.$$

Since $\text{Tr}(\tilde{B}_\nu E_\mu) = r_\mu x_\nu + s_\mu z_\nu + t_\mu y_\nu + u_\mu w_\nu$, put $U = \mathbf{r} \otimes \mathbf{x} + \mathbf{s} \otimes \mathbf{z}$ and $\mathcal{V} = \mathbf{t} \otimes \mathbf{y} + \mathbf{u} \otimes \mathbf{w}$, so $4F' = \|U + \mathcal{V}\|^2$. The cross terms carry the vanishing factors $\langle \mathbf{r}, \mathbf{s} \rangle$, $\langle \mathbf{t}, \mathbf{u} \rangle$, giving

$$\|U\|^2 = a\|\mathbf{x}\|^2 + b\|\mathbf{z}\|^2 \leq a, \quad \|\mathcal{V}\|^2 = (1-a)\|\mathbf{y}\|^2 + (1-b)\|\mathbf{w}\|^2 \leq \eta(1-b);$$

the triangle inequality then yields the **covariance-free coupled bound**

$$4F' \leq (\sqrt{a} + \sqrt{\eta(1-b)})^2. \quad (*)$$

This uses no covariance, unitality, reality, or invertibility.

Label constraint. Since σ_d is a state, $\text{Tr}(A\sigma_d) \geq \alpha\eta$, and with $p = \sigma_s[0,0] = (a+b)/2$ one gets $L \leq \frac{1}{2} + \frac{1}{2}\alpha(1-\eta)^{\frac{a+b}{2}}$; hence, with $\kappa := 4\tau - 2 \in [0, 2]$,

$$L \geq \tau \implies \alpha(1-\eta)(a+b) \geq \kappa. \quad (\dagger)$$

Reduced scalar program and certificates. For $0 < \kappa \leq 2$, maximizing the envelope $O = \frac{\alpha}{4}(\sqrt{a} + \sqrt{\eta(1-b)})^2$ subject to $(\dagger)$ is a relaxation of the channel problem (realizability is supplied in Section 5). Because O and the left side of $(\dagger)$ are both increasing in a and α , setting $\alpha = a = 1$ is optimal. Writing $u = 1 - \eta \in [0, 1]$, the constraint $(\dagger)$ reads $u(1+b) \geq \kappa$ with $b \in [0, 1]$, so feasibility is $u \geq \frac{\kappa}{2}$ with $b_{\min}(u) = \max\{0, \frac{\kappa}{u} - 1\}$. The coherence term is $q(u) = 1 - u$ on the *slack* branch ($u \geq \kappa$, $b = 0$, present only when $\kappa \leq 1$) and $q(u) = g(u) := 2 + \kappa - 2u - \kappa/u$ on the *active* branch $\frac{\kappa}{2} \leq u \leq \min\{\kappa, 1\}$ (here $u \geq \frac{\kappa}{2} > 0$, so κ/u is defined). Exact nonnegative residual identities certify the maxima without any search:

$$0 < \kappa \leq \frac{1}{2} : \quad (1 - \kappa) - g(u) = \frac{(\kappa - u)(1 - 2u)}{u} \geq 0, \quad (1 - \kappa) - (1 - u) = u - \kappa \geq 0;$$

$$\frac{1}{2} \leq \kappa \leq 2 \ (\kappa = 2h^2) : \quad 2(1 - h)^2 - g(u) = \frac{2(u - h)^2}{u} \geq 0, \quad 2(1 - h)^2 - (1 - u) = (2h - 1)^2 + u - \kappa \geq 0.$$

(These are products or affine expressions with a manifest sign on the stated domains, not all pure squares.) Hence $\max_u q = 1 - \kappa$ for $\kappa \leq \frac{1}{2}$ and $\max_u q = (\sqrt{2} - \sqrt{\kappa})^2$ for $\kappa \geq \frac{1}{2}$, so

$$f_{\text{all}}(\tau) \leq \Phi(\tau) = \frac{1}{4}(1 + \sqrt{\max_u q})^2.$$

The three boundary cases are discharged directly. For $\kappa = 0$ ($\tau = \frac{1}{2}$), $(\dagger)$ is vacuous and $F_{\text{same}} \leq 1$ by definition, attained by the perfect-code strategy $(\gamma, \eta) = (0, 1)$. For $\kappa = 2$ ($\tau = 1$), $(\dagger)$ forces $\alpha(1 - \eta)(a + b) \geq 2$. With $0 \leq \alpha \leq 1$, $0 \leq 1 - \eta \leq 1$, $0 \leq a + b \leq 2$, the exact identity

$$2 - \alpha(1 - \eta)(a + b) = 2(1 - \alpha) + 2\alpha\eta + \alpha(1 - \eta)[(1 - a) + (1 - b)]$$

has all summands nonnegative on the cube, so $\alpha(1 - \eta)(a + b) \leq 2$ with equality only when $\alpha = 1$, $\eta = 0$, $a = b = 1$; then $(*)$ gives $F_{\text{same}} \leq \frac{1}{4}$, attained by the family at $(\gamma, \eta) = (1, 0)$. The zero-effect case $A = 0$ was disposed of above. This is the two-branch formula of Theorem 1. $\square$

5. An explicit covariant family that attains the bound

The bound is achieved by a two-parameter covariant CPTP family. The **encoder** applies covariant amplitude damping of strength $\gamma \in [0, 1]$ to the code (in the Hadamard-rotated memory basis, which sends logical X to Z , so the counted qubit carries $V = Z$) and routes the different sector to a classical flag; the **decoder** applies a single covariant same-sector recovery with branch effect $\text{diag}(1, \eta)$, $\eta \in [0, 1]$, and completes on Q_d . Explicitly, with H the 2×2 Hadamard,

$$\mathcal{E}_\gamma : \left\{ \text{diag}(1, \sqrt{1 - \gamma}) HW^\dagger, \quad \sqrt{\gamma} |0\rangle\langle 1| HW^\dagger, \quad |1\rangle\langle 01|, \quad |1\rangle\langle 10| \right\},$$

$$\mathcal{D}_\eta : \left\{ WH \text{diag}(1, \sqrt{\eta}), \quad \sqrt{\frac{1 - \eta}{2}} |01\rangle\langle 1|, \quad \sqrt{\frac{1 - \eta}{2}} |10\rangle\langle 1| \right\}.$$

Both factors are CPTP and factorwise covariant with memory representation Z for all $(\gamma, \eta) \in [0, 1]^2$ (verified as polynomial identities on every matrix unit), realize $a = 1$, $b = \gamma$, $\alpha = 1$, and have closed-form scores

$$L(\gamma, \eta) = \frac{1}{2} + \frac{(1 - \eta)(1 + \gamma)}{4}, \quad F_{\text{same}}(\gamma, \eta) = \frac{1}{4}(1 + \sqrt{\eta(1 - \gamma)})^2.$$

Maximizing F_{same} over the family at $L \geq \tau$ is *exactly* the reduced program of Section 4, so its optimum is $\Phi(\tau)$, attained at

$$\tau \in [\frac{1}{2}, \frac{5}{8}] : \gamma_\star = 0, \eta_\star = 3 - 4\tau; \quad \tau \in [\frac{5}{8}, 1] : \gamma_\star = 2\sqrt{2\tau - 1} - 1, \eta_\star = 1 - \sqrt{2\tau - 1}.$$

Substituting the optimizers returns $(L, F_{\text{same}}) = (\tau, \Phi(\tau))$ as a polynomial identity on each branch. This gives $f_{\text{cov}}(\tau) \geq \Phi(\tau)$, completing Theorem 1.

6. Structure of the frontier

A transition at $\tau = \frac{5}{8}$ (a necessity). Consider the code-restricted encoder T (the full encoder $\mathcal{E} : \mathbb{C}^4 \rightarrow \mathbb{C}^2$ can never be unitary). If T is unitary then $G = T^\dagger(|0\rangle\langle 0|)$ has eigenvalues $a = 1, b = 0$, and $(*)-(\dagger)$ give the unitary-encoder optimum $(1 + \sqrt{1 - \kappa})^2/4 = (1 + \sqrt{3 - 4\tau})^2/4$, which requires $\kappa \leq 1$, i.e. no unitary T exists above $\tau = \frac{3}{4}$. For $\frac{5}{8} < \tau \leq \frac{3}{4}$ the full optimum strictly exceeds it: with $\kappa = 2h^2$,

$$2(1 - h)^2 - (1 - 2h^2) = (2h - 1)^2 > 0 \quad (h \neq \frac{1}{2}).$$

Hence **at or below $\frac{5}{8}$ a unitary code encoder can be optimal, and strictly above $\frac{5}{8}$ every optimal code-restricted encoder is nonunitary.** This is a statement about *all* optimizers (necessity of nonunitarity); it is not a claim that every optimizer is the specific amplitude-damping map of Section 5, which would require a further equality-case classification. The two regimes join C^1 at $\frac{5}{8}$.

Nonconvexity of the attainable region. The covariant endpoint strategies attain $(\frac{1}{2}, 1)$ and $(1, \frac{1}{4})$, both in both classes, yet their midpoint $(\frac{3}{4}, \frac{5}{8})$ exceeds the proven optimum $\Phi(\frac{3}{4}) = \frac{1}{2}$ and so lies in *neither* attainable region. Thus each fixed-resource attainable (L, F_{same}) region is **nonconvex**. This is why the prohibition of an uncoupled shared classical index between $\mathcal{E}$ and $\mathcal{D}$ is mathematically consequential rather than mere bookkeeping: were correlated encoder–decoder mixing free, the region would be convex and the midpoint attainable.

7. Related work

The task sits near two established families, and we position it against both without claiming novelty. Throughout, we keep “a specific reduction does not go through” strictly separate from “no reduction exists,” and we never argue from the proved frontier to a predecessor.

Binary discrimination versus disturbance (Buscemi–Sacchi). Buscemi and Sacchi (quant-ph/0610196) characterize the optimal trade-off between the success probability of discriminating two fixed nonorthogonal pure states and an operation fidelity averaged over those same two states, and give an instrument-symmetrization argument (their Eq. 16) preserving both scores. Since $L = P$ is a discrimination success probability, the information axes align, and sub-optimal (sub-Helstrom) discrimination is allowed in their setting, so it is not an obstruction to comparison. What differs is structural: their two states are prescribed *inputs*, whereas ours are variable encoder *outputs* σ_s, σ_d (with σ_s of encoder-dependent rank and σ_d pure in the displayed cases); their disturbance is an average over two states, ours an asymmetric single-sector entanglement fidelity; and their instrument carries no counted-memory stage. Their symmetrization preserves *their* scores but does not by itself establish our counted-memory, factorwise-covariant attainability. The relation is that of a genuine information–disturbance cousin whose naive reduction is not established here.

One-shot and hybrid classical–quantum coding (Kuperberg; Salek et al.; Devetak–Shor). Writing $\mathcal{N} = \mathcal{D} \circ \text{id}_2 \circ \mathcal{E}$ exhibits the task as one use of a noiseless qubit with variable encoder

and decoder. Appendix A recasts it exactly as storing a hybrid memory $M_2 \oplus \mathbb{C}$ (a “hybrid trit,” Kuperberg shape $(2, 1)$) through a qubit (shape (2)). This is precisely the setting of Kuperberg’s one-shot Hölder-type fidelity inequality (quant-ph/0203105, Thm. 4.1), whose hypothesis — that the storage maps be “decay” (subunitary CP) — is met here, since a CPTP map has a unital, hence subunitary, CP adjoint. Specializing the inequality (equivalently, a self-contained argument from the factorization: the superoperator-trace identity $\text{Tr}_{\text{sup}}(\mathcal{N}_{\mathcal{H}}) = 4F_{\text{same}} + t_d$, rectangular cyclicity, and that the reversed composition is a CPTP qubit channel with Kraus $\{K_j\}$ so that $\sum_j |\text{Tr } K_j|^2 \leq 2 \sum_j \text{Tr}(K_j^\dagger K_j) = 4$) gives $4F_{\text{same}} + t_d \leq 4$; with $t_s \leq 1$ and $L \geq \tau$, so $t_d \geq 2\tau - 1$,

$$F_{\text{same}} \leq \frac{5-2\tau}{4}.$$

Concretely, Thm. 4.1 with $\rho = (\frac{4}{5} \cdot \frac{1}{2}) \oplus \frac{1}{5}$, $p = 1$, $q = \infty$ has $\|\rho\|_d = \frac{1}{5}$ and $\|\lambda(2, 1)\|_\infty \|\lambda(2)\|_1 = 4$, i.e. $\frac{1}{5}(4F_{\text{same}} + t_d) \leq \frac{4}{5}$, the same inequality. This bound is *valid but strictly loose* in our application: it is tight only at $\tau = \frac{1}{2}$ (both sides 1) and gives $\frac{7}{8}$ against the sharp $\Phi(\frac{3}{4}) = \frac{1}{2}$. Salek et al. (arXiv:1809.07104) give one-shot achievability and converse bounds whose Definition 18 and Theorems 2–3 concern public–private coding (average joint decoding error with a trace-distance secrecy condition), with the classical–quantum reading developed separately; translating that to our single-sector quantum criterion is not carried out here. Devetak–Shor (quant-ph/0311131) give the asymptotic simultaneous region. In sum, a directly applicable prior bound exists and is loose; whether a sharper prior derivation recovers Φ , the transition, or covariant optimality is open.

Fidelity balance. Banaszek’s fidelity-balance bounds (quant-ph/0003123) trade estimation fidelity against state disturbance for a single system; the differing task (no counted memory, no sector label, symmetric fidelity) makes them a background reference rather than a reduction.

8. Scope and discussion

The result is an equality of *optima at each threshold*, at the memory representation $V = Z$: the best same-sector fidelity at a required label accuracy is the same for covariant and unrestricted single-qubit factorizations. It does not assert that the two attainable regions coincide off the frontier, nor that every unrestricted channel has a covariant counterpart matching both scores; and it concerns a single unassisted qubit factorization rather than an arbitrary effective channel or a shared-randomness-assisted mixture (the composite $\mathcal{N}$ is itself an effective channel — it is the qubit-factorability, unassisted, that is the restriction). Within that scope the conclusion is definite and structural: on this task, at $V = Z$, respecting the $\mathbb{Z}_2$ symmetry never costs optimal same-sector fidelity, at any reliability of the classical record; the nonunitarity transition and the nonconvexity of the region are exact features of the frontier; and a directly applicable hybrid-memory bound is weaker here than the sharp result. Any distinguishing effect of the symmetry on this task must therefore be sought somewhere other than an optimal-fidelity penalty on this trade-off. We make no claim beyond the mathematics stated here; interpretive applications are developed separately and are not part of this result.

Appendix A. An exact hybrid-memory reformulation

The task is equivalent to a channel on the hybrid algebra $\mathcal{H} = M_2 \oplus \mathbb{C}$ (a qubit block and a classical flag) through the boundary maps

$$\iota(R \oplus p) = WRW^\dagger + p \frac{Q_d}{2}, \quad \pi(Y) = W^\dagger Y W \oplus \text{Tr}(Q_d Y).$$

These are CPTP with $\pi \circ \iota = \text{id}_{\mathcal{H}}$; the test inputs $(\frac{1}{2}) \oplus 0$ and $0 \oplus 1$ map to $Q_s/2$ and $Q_d/2$; and for any admissible $\mathcal{N}$, the hybrid channel $\mathcal{N}_{\mathcal{H}} = \pi \circ \mathcal{N} \circ \iota$ preserves both scores and factors through the same counted qubit, with the symmetry $U_{\mathcal{H}} = X \oplus 1$ intertwining ι, π with $U = X \otimes X$ and the qubit carrying $V = Z$. This is the identification used in Section 7: $\mathcal{H}$ is Kuperberg’s hybrid trit (shape $(2, 1)$), and the counted qubit is shape (2) .

Conversely, any admissible *hybrid* channel lifts to the carrier. Given hybrid factors $\mathcal{E}_{\mathcal{H}} : \mathcal{H} \rightarrow \mathcal{L}(\mathbb{C}^2)$ and $\mathcal{D}_{\mathcal{H}} : \mathcal{L}(\mathbb{C}^2) \rightarrow \mathcal{H}$, the maps $\mathcal{E} = \mathcal{E}_{\mathcal{H}} \circ \pi$ and $\mathcal{D} = \iota \circ \mathcal{D}_{\mathcal{H}}$ are CPTP, use the same counted qubit, and give a carrier channel with identical L and F_{same} ; and if $\mathcal{E}_{\mathcal{H}}, \mathcal{D}_{\mathcal{H}}$ intertwine $\text{Ad}_{X \oplus 1}$ with Ad_Z , then $\mathcal{E}, \mathcal{D}$ are carrier-covariant with $V = Z$. Hence the two formulations have the *same attainable score pairs* in their corresponding classes, by these constructions; an explicit lift is checked in the regression script. This is an equality of attainable scores, not a bijection of all effective channels: $\iota \circ \pi$ is not the identity on the full carrier algebra $\mathcal{L}(\mathbb{C}^4)$.

Appendix B. Reproducibility

The results are proved analytically in the sections above; the analytic proofs are the primary evidence. Independently, the following exact symbolic computations (SymPy, no floating point in the certified checks) verify selected identities, explicit constructions, and regression examples; each script exits nonzero if any assertion fails. These seven verification scripts, a README, and a runner are provided as the supplementary archive `coexistence_verification_supplement.zip`, archived at Zenodo, DOI [10.5281/zenodo.23049858](https://doi.org/10.5281/zenodo.23049858):

- `extend_tradeoff.py`
- `audit_closure.py`
- `verify_hybrid_dictionary.py`
- `verify_kuperberg_specialization.py`
- `verify_manuscript_regressions.py`
- `comparisons_checks.py`
- `q2_reduction.py`

The environment used was Python 3.11.15, SymPy 1.14.0, NumPy 2.4.4 (NumPy is used only by the separately labelled floating-point corroboration in the first two scripts). To reproduce, unpack the archive into an empty directory and run `bash run_all.sh` (or `python3 <script>.py` for each), confirming the ALL ... PASS footer and zero exit status. Figure regeneration is a separate, optional step (`make_tradeoff_fig_ms.py`, requiring Matplotlib) and is not part of the verification entry point. The randomized floating-point sweeps that two scripts print are separately labelled corroboration, not part of the certified counts.

The frontier and attaining family (`extend_tradeoff.py`) check the coupled bound’s monotonicity reductions, the domain bookkeeping and the four nonnegative residual identities (products or affine expressions with a manifest sign, not all pure squares), the family’s CPTP property and covariance on every matrix unit, its closed-form scores, the optimizer match $(\tau, \Phi(\tau))$ as a polynomial identity on each branch, the C^1 match and transition-gap identity $(2h - 1)^2 > 0$, and the nonconvexity midpoint (32 exact assertions, with the $\kappa = 0/A = 0$ endpoints and optimizer feasibility argued analytically and labelled as such). The closure point $f_{\text{cov}}(\frac{3}{4}) = f_{\text{all}}(\frac{3}{4}) = \frac{1}{2}$ is audited exactly in `audit_closure.py` (Hermitian norm-loss identities for arbitrary complex Kraus, the square certificate as a structural perfect-square identity, and the achiever saturating every step). The identity $L = P$ and the hybrid dictionary of Appendix A are reproduced in `verify_hybrid_dictionary.py` (43 exact checks; the counted qubit carries the Z sign class, and

for the *displayed* $\tau = \frac{3}{4}$ *achiever in its basis* the trivial and other Pauli representations fail to intertwine — this is a property of that example, not of every threshold: the endpoint optimizer $(\gamma, \eta) = (1, 0)$ is covariant for both Z and $\mathbb{1}$). The Kuperberg specialization $F_{\text{same}} \leq (5 - 2\tau)/4$ is verified in `verify_kuperberg_specialization.py` (31 exact checks: the residual decomposition into explicit real squares each provably nonnegative, the superoperator-trace bookkeeping and rectangular cyclicity, the whole-interval gap identities $B - \Phi = (1 - x)(x + 5)/8$ and its upper-branch analogue with manifestly positive coefficients, and the explicit $p = 1, q = \infty$ substitution). The consolidation examples of this revision — the sector action, the label-swap with $L = 0$, the zero-effect $(\frac{1}{2}, 0)$ channel, the $V = \mathbb{1}$ achiever $(\frac{1}{2}, \frac{1}{2})$, the endpoint’s dual-representation covariance, the $\kappa = 2$ corner (as an exact residual identity), and the reverse dictionary lift (on one specific channel) — are checked in `verify_manuscript_regressions.py` (15 exact checks). The resource/score facts used in Section 7 are checked in `comparisons_checks.py` (9), and the certified failure of the whole-channel twirl as a covariance reduction is recorded with an exact rank witness in `q2_reduction.py` (23).

Code and data availability

This work — this manuscript together with its verification supplement (Appendix B: seven exact-computation scripts, a README, and a runner) — is archived at Zenodo, DOI [10.5281/zenodo.23049858](https://doi.org/10.5281/zenodo.23049858). The manuscript uses no experimental data.

Acknowledgments and disclosure

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